

Deploy secure digital product passports for electric vehicle battery replacements to meet new EU authorization procedures.

Automotive × Regulatory compliance and reporting × Post-quantum cryptography · OS067 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
31 is the world moving	42 can we play, can we win	MEDIUM 0 named in the evidence	€225m modelled evidence · per year	NEXT regulation adopted or propose...	L1 4 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity addresses the EU's new authorization procedure for replacement rechargeable electric energy storage systems (REESS) in motor vehicles, which creates a compliance need for automotive manufacturers and their supply chains. By deploying secure digital product passports, Orange Business can help customers meet these regulatory requirements while leveraging its expertise in cybersecurity and digital transformation. The time to act is now, as the regulation is adopted but not yet applicable, giving early movers a competitive advantage.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€254m €47.5m – €1.2bn	€225m €42.1m – €1.1bn	€2.7m €505k – €12.8m	modelled
Observed: tendered contract value, annualised	€10.0m	€10.0m	€120k	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Automotive (EU27_2020)	6,364 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Post-quantum cryptography	49.0 % of enterprises (10+ employees)	proxy	Eurostat · isoc_cisce_ran2 · E_SECMDENC	2024
Annual value of one engagement	€260k per enterprise per year	assumption	config/sizing.yaml · fallback_bands_eur	size-2026-08-a
Size-weighted buyer base	1,995 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. ICT security measures by NACE Rev. 2 activity series E_SECMDENC is used as a proxy for Post-quantum cryptography: Encryption users are the migration population. Contract value is a configured assumption, not observed. Only 1 matching tender notices, below the 6 needed for a robust median, so the configured band for this business domain is used. It is an assumption owned by innovation-radar-curator, not evidence. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

Observed: tendered contract value, annualised

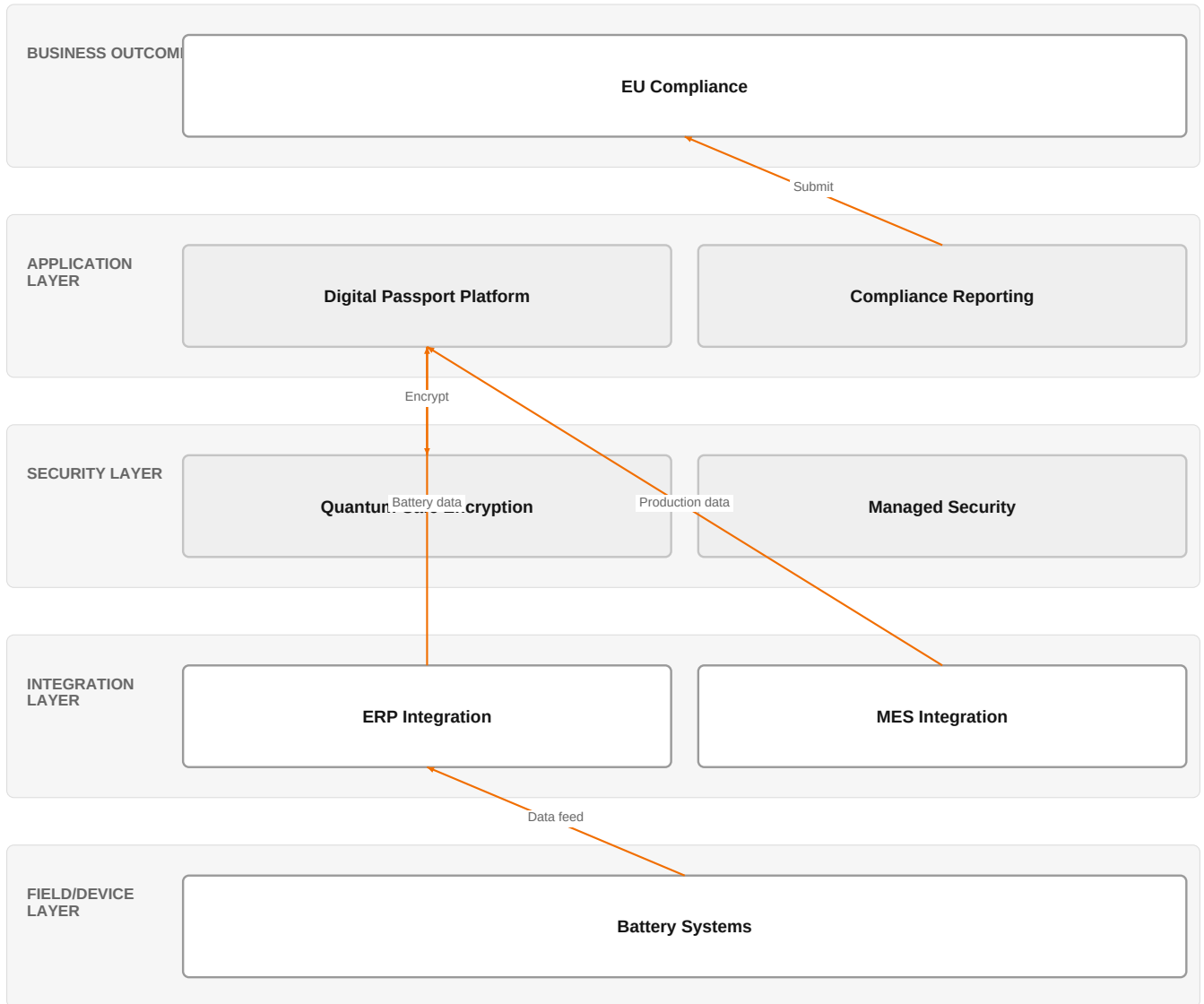
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€10.0m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 6 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in

this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

Secure Battery Passport Solution



Orange asset Partner Customer-owned Third party / to be sourced

This diagram shows how Orange's secure digital passport platform integrates with existing customer systems to ensure EU compliance.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The EU has established a procedure for authorizing replacement REESS, mandating compliance for automotive players. Digital product passports are emerging as a key tool for circularity and transparency in battery lifecycle management, as highlighted in recent research. The adoption of asset administration shells and digital passports in MRO processes signals a broader industry shift toward data-driven, autonomous maintenance. This regulatory and technological convergence creates a pressing need for secure, compliant digital solutions.

Sources: SIG-56CA6A58B3BF ec.europa.eu 2026-02-03; SIG-41FB7CD0130C openalex.org 2025-10-21; SIG-E6157B37F004 openalex.org 2025-09-14; SIG-03A707688978 Automotive IQ 2026-05-19

Who buys, and why now

The primary buyer is the Chief Compliance Officer or Head of Regulatory Affairs at an automotive OEM or major battery supplier. The pain is operational: they must track and report on battery replacements in a way that satisfies the EU authorization procedure, which is currently manual and error-prone. The trigger is the upcoming applicability of the regulation, forcing them to implement a compliant process or face market access barriers. The IT department feels the

pain of integrating new data requirements into existing systems.

Sources: SIG-56CA6A58B3BF ec.europa.eu 2026-02-03

Why it is hot now — every claim bound to a dated source

- The EU has established a procedure for authorizing replacement rechargeable electric energy storage systems in motor vehicles, creating a compliance need. [SIG-56CA6A58B3BF]
- The adoption of digital product passports and asset administration shells is advancing maintenance processes in Industry 4.0. [SIG-41FB7CD0130C]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a secure digital product passport solution leveraging its Cybersecurity experts and Digital experts. The engagement would start with a compliance assessment to map the customer's current battery data flows against the EU procedure. Then, Orange would design and implement a digital passport platform, integrating with the customer's existing ERP and MES systems, and secure it with quantum-safe encryption using Quantum Defender / quantum-safe offers. Finally, Orange Cyberdefense managed services would provide ongoing monitoring and threat detection.

Why Orange can win

Orange can win by combining its cybersecurity expertise with its digital transformation capabilities, offering a secure-by-design solution that addresses the trust and sovereignty requirements of the automotive sector. The TISAX certification demonstrates Orange's commitment to automotive security standards, which is a key differentiator. However, the asset list is thin on specific automotive domain expertise, so Orange must leverage its reference with Toyota to build credibility.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Quantum Defender / quantum-safe offers	offer	L1	offer provides the technology in a matching domain	unconfirmed
Orange Cyberdefense managed services	offer	L1	offer addresses the use case but not this technology	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Cybersecurity experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Healthcare experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Toyota	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 6.4; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
Atos / Eviden	Global systems integrator	sells Post-quantum cryptography; competes in Cloud, Cybersecurity	structural	—
IBM	Global systems integrator	sells Post-quantum cryptography; competes in Cloud, Cybersecurity	structural	—

Airbus Protect	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
Entrust	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
ID Quantique	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
Thales	Security specialist	sells Post-quantum cryptography; competes in Cloud, Cybersecurity	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	active in Automotive; competes in Cloud, Cybersecurity	structural	—
Vodafone Business	Telecom operator peer	active in Automotive	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How are you currently tracking and documenting battery replacements for vehicles in the EU?
- 2 What is your current process for ensuring compliance with the new EU authorization procedure for replacement REESS?
- 3 Do you have a digital product passport initiative in place, and if so, what stage is it at?
- 4 Who in your organization is responsible for regulatory compliance for battery systems?
- 5 What are your main concerns about meeting the EU's data requirements for battery traceability?
- 6 Are you already using any post-quantum cryptography or quantum-safe solutions in your IT infrastructure?

Objections you will meet

They will say	You can answer
We already have a battery passport project with another vendor.	That's a common situation. However, our solution is designed to be complementary, focusing on the security and compliance aspects that are often overlooked. We can integrate with your existing systems and add quantum-safe protection to ensure long-term data integrity.
The regulation is not yet applicable, so we have time.	True, but early movers will have a smoother transition and can influence industry standards. Starting now allows you to pilot and refine your processes before the deadline, reducing last-minute risks and costs.
We are concerned about the cost and complexity of implementation.	We understand. Our approach is modular, starting with a compliance assessment to identify the minimal viable changes. We can phase the implementation to align with your budget and operational constraints, leveraging our managed services to reduce upfront investment.

Risks and unknowns

The main risk is that the regulation's applicability date may shift, reducing urgency. There is also uncertainty about the specific data requirements for the digital passport, which could change as the EU finalizes implementation details. Additionally, the integration complexity with legacy automotive systems is unknown and could cause delays. Finally, the competitive landscape is crowded, and Orange's lack of a dedicated automotive vertical practice (beyond TISAX) may be a disadvantage.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a European automotive OEM, highlighting Orange's TISAX certification, Quantum Defender, and managed services to support secure digital product passports for EV battery replacements, referencing the Toyota engagement as proof of capability.
Sales	In a conversation with a European automotive manufacturer's head of compliance, mention Orange's Quantum Defender and managed cybersecurity services as a way to future-proof their digital product passport implementation against quantum threats.
Strategist / Innovator	Commission a deep-dive brief on how post-quantum cryptography can be integrated into digital product passports for EV battery replacements, leveraging Orange's Quantum Defender and cybersecurity experts to address EU authorization compliance.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-56CA6A58B3BF	2026-02-03	ec.europa.eu	1	Procedure for authorisation of replacement rechargeable electric energy storage systems (REESS)...
SIG-41FB7CD0130C	2025-10-21	openalex.org	1	Advanced MRO Processes in Industry 4.0 with Proactive Asset Administration Shell and Digital Pr...
SIG-E6157B37F004	2025-09-14	openalex.org	1	Advancing Circularity in Battery Systems for Renewable Energy: Technologies, Barriers, and Futu...
SIG-03A707688978	2026-05-19	Automotive IQ	2	The EU Battery Passport Explained: Requirements, Timeline and Compliance Steps for Automotive]...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (4 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:56:42+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.